

OTS06: FAQs

How much historical data do I need to get statistically significant results for my options strategy?

A: For monthly strategies, you typically need at least 2-3 years of data for basic validation, but 5-10 years is preferable for robust statistical significance. For weekly strategies, 1-2 years might suffice since you'll have more trades. Consider market cycle coverage - ensure your data includes bull, bear, and sideways markets. The key is having enough independent trades rather than just a time duration.

How do I backtest strategies that require real-time Greeks when I only have end-of-day options data?

A: You have three approaches:

- (1) Calculate theoretical Greeks using Black-Scholes with market IV at each timestamp,
- (2) Use historical Greeks if available in your dataset, or
- (3) Interpolate Greeks between available data points.

For delta-hedging strategies, you may need intraday data since delta changes continuously. Consider whether your strategy truly requires real-time Greeks or if daily rebalancing is sufficient.

Should I use bid prices, ask prices, or mid prices for entry and exit in my backtester?

A: Use bid prices for selling and ask prices for buying to be realistic about execution. For small retail quantities, you might assume mid-price execution with a fixed spread penalty. For larger institutional strategies, model market impact. Never use last traded prices unless you have supporting volume data indicating realistic execution possibility.

How do I backtest strategies that involve early exercise of American options?

A: Implement early exercise logic based on theoretical conditions (deep ITM puts before dividends, calls on dividend dates). Most retail backtesting can ignore early exercise since it's rare, but for short options strategies or dividend capture strategies, it's crucial. Consider using Monte Carlo methods for complex early exercise scenarios.

My strategy shows great returns but a terrible Sharpe ratio. What does this mean?

A: High returns with a low Sharpe ratio indicate high volatility or significant drawdowns. This is common with options strategies that collect premiums regularly but occasionally suffer large losses. Consider position sizing, risk management rules, or strategy modifications to smooth returns. Remember that the Sharpe ratio assumes normal return distributions, which options strategies often violate.

What's the best way to handle partial fills in options backtesting?

A: For backtesting purposes, assume full fills for liquid options meeting your filter criteria. If modelling partial fills, use historical volume data to estimate fill probability. For professional backtesting, implement order book simulation, but this requires market depth data. Most academic and retail backtesting assumes perfect execution at filtered prices.

How do I incorporate margin requirements into my options backtester?

A: Calculate margin requirements based on your broker's rules (typically SPAN margining for portfolios). Track available buying power throughout the backtest and prevent new positions if insufficient margin. For cash-secured puts, reserve the full cash amount. For covered calls, own the underlying stock. Include margin interest costs in your P&L calculations.

Should I model slippage as a fixed percentage or dynamically based on market conditions?

Start with a fixed percentage (0.25-0.5% of premium) for simplicity, but dynamic slippage is more realistic. Consider higher slippage during volatile periods, for illiquid options, or large orders. You could model slippage as a function of bid-ask spread, volume, or VIX level. Document your assumptions clearly since slippage modelling significantly impacts results.

My strategy performs well in backtesting but fails in paper trading. What went wrong?

A: Common issues include:

- (1) Look-ahead bias in signals
- (2) Unrealistic execution assumptions
- (3) Missing transaction costs
- (4) Data-snooping bias from over-optimisation
- (5) Regime changes in market structure

(6) Insufficient out-of-sample testing.

Always test on truly unseen data and implement the strategy exactly as backtested.

How do I account for the changing options landscape (more weekly options, smaller spreads, electronic trading)?

A: Use data that reflects the time period you plan to trade. Recent historical data (last 3-5 years) better represents the current market structure than older data. Adjust transaction cost assumptions based on current broker fees. Consider that strategies profitable in older, less liquid markets might not work in today's competitive environment.

How do I backtest multi-leg strategies with complex exit rules (like iron condors with rolling)?

A: Break down complex strategies into individual legs, track each leg separately, then aggregate P&L. Implement a logic to handle different exit conditions (profit target, loss limit, time decay, delta threshold). Use clear naming conventions for position tracking and implement robust position reconciliation checks.

What's the best way to parallelise options backtesting for faster execution?

A: Parallelise by time periods (different years/months) or by strategy variations rather than individual calculations, since options pricing has path dependencies. Use vectorised operations in pandas/numpy for speed. Consider caching computed Greeks and other derived values. Profile your code to identify bottlenecks - often, data I/O is the limitation, not computation.

How do I handle missing data points in my options chains?

A: For missing strikes, you can interpolate using surrounding strikes, but be cautious about introducing artificial patterns. For missing time series points, either skip those dates or use forward-fill carefully. Document all data handling decisions. Sometimes missing data indicates genuine illiquidity, which is valuable information itself.